

Project Design Phase-I Proposed Solution Template

Date	June 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	AI-Enhanced Intrusion Detection System
Maximum Marks	As Applicable

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Traditional Intrusion Detection Systems (IDS) rely on signature-based rules and fail to detect novel/zero-day attacks. Security analysts face high false positive rates (~60%), wasting hours on manual log analysis. There is no intelligent, real-time classification of attack types, leaving organizations vulnerable to DoS, Probe, R2L, and U2R attacks.
2.	Idea / Solution Description	An AI-Enhanced Intrusion Detection System trained on the NSL-KDD dataset using a Random Forest classifier. The system automatically classifies network traffic into 5 categories (Normal, DoS, Probe, R2L, U2R), exposed via a Flask REST API, and visualized on a real-time SOC dashboard built with HTML, CSS, and Chart.js.
3.	Novelty / Uniqueness	Unlike rule-based IDS tools (e.g., Snort), this system uses machine learning to detect previously unseen attack patterns. It provides real-time threat classification with confidence scores, reducing false positives to under 5% and enabling proactive, data-driven incident response.
4.	Social Impact / Customer Satisfaction	Reduces cybersecurity breach risk for organizations of all sizes. Security analysts spend less time on false alarms and more time on genuine threats. Faster incident response protects sensitive data, builds stakeholder trust, and reduces financial and reputational damage from cyberattacks.
5.	Business Model (Revenue Model)	SaaS subscription model: tiered pricing for SMEs, enterprises, and government organizations. Free tier for basic monitoring; paid tiers for real-time analytics, custom alerting, and API integrations. Additional revenue from professional services and cybersecurity consulting.
6.	Scalability of the Solution	The Flask API can be containerized with Docker and deployed on cloud platforms (AWS/Azure/GCP). The ML model can be retrained periodically with new data. The dashboard supports multiple simultaneous analyst sessions.

		The architecture supports horizontal scaling to handle millions of network events per day.
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